Sailfort Motors' Employee Retention Identification Model

Prepared for: Sailfort Motors HR Team

Project Overview

Sailfort Motors seeks to reduce the rate of employees leaving the company. The HR team needs a model that is capable of accurately predicting whether an employee will stay with the company or leave, so they can identify factors contributing to employees leaving. They will use this information to implement interventions to improve employee retention. This will lead to cost and time savings as the company invests heavily in training new employees.

Key Insights

1) An initial exploratory data analysis revealed the **general profiles of employees most at risk of leaving the company**. These include employees who:

- ➢ Feel unsatisfied, overworked, and under-appreciated due to heavy work loads and no recent promotions.
- ➢ Feel satisfied and perform well at their work, but work very long hours.
- ➢ Have 6 or more projects.
- ➢ Have only 2 projects.
- ➢ Feel unsatisfied and perform poorly at their jobs.

2) A Chi-Square test determined that employee department had little or no impact on the rate employees left the company. **No employee in one department is statistically more likely to leave than any other department.** Due to this and a heightened risk of overfitting, this variable was not used in model development. To ensure the model is able to be used in the future without subjective employee input, satisfaction score was dropped also.

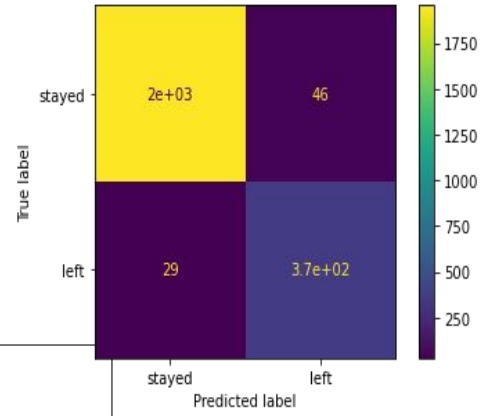
3) Three models were developed, including a logistic regression and two tree-based machine learning models. After comparing all three using validation data, **the champion model was the random forest machine learning model**. Additional testing using novel test data revealed the model is excellent at correctly identifying both employees who will stay with the company and those who will leave. In the future, the model is **expected to correctly predict 96.9% of employees who will be retained and 92.7% of employees who will leave**.

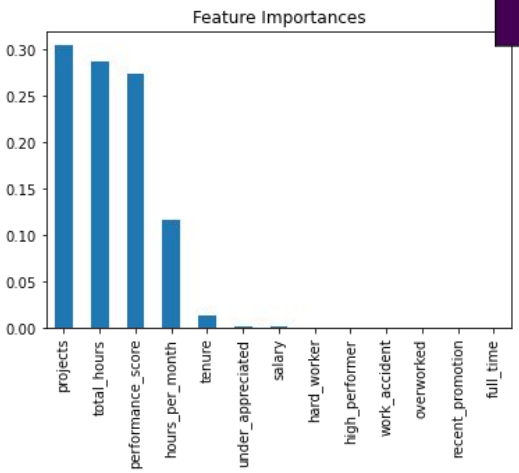
4) **The most important factors contributing to the model are projects, total hours worked for the company, recent performance score, and average hours worked per month.** This aligns with the general employee profiles developed during the exploratory analysis.

Details

Model Performance

Only 29 employees who left were missed by the model after predicting about 10,000 people. ***Employees who left were correctly identified 92.7% of the time.***





Contributing Factors

Employees who worked 6 or more projects and longer hours per month were more likely to leave the company. *Projects was found to be the most important feature used by the model.*

Next Steps

- *Evidence suggests that there is not a major issue among specific departments affecting employee retention rate. Interventions can be implemented more broadly among the employee population rather than tailoring the approach to select departments.*
- *Implement a policy to cap the amount of projects employees can work on at one time to 5. Between 3 and 5 projects at a time would be ideal for most employees.*
- *Data indicates employee burnout is pervasive throughout the company due to a culture of long hours and heavy workloads. Ensure benefit packages are sufficient and investigate whether employees have confidential access and awareness of mental health care, such as online therapy.*
- *Employee burnout is a common issue in issue in corporate workplaces. Research ways other companies have successfully addressed this issue.*